LLMOps for Drug Discovery Systems

Governing Models, Agents, and Evaluation

Foundation models and AI agents are transforming drug discovery, but real-world deployment demands strong MLOps and LLMOps. Using Weights & Biases Models and Weave, we demonstrate a reproducible workflow for fine-tuning, agent tracing, evaluation, and RL that improves reliability, speeds iteration, and supports compliance.

BioIT World • 22-minute talk

Talk Structure (22 min)

1. **Case for a systematic improvement loop** (6 min)
2. **W&B tools and what they unlock** (5 min)
3. **Our BioReason-Pro plan: ablations + tuning strategies** (6 min)
4. **Results + live W&B demo** (5 min)

Slides for narrative + live UI for evidence.

1) Why We Need a Systematic Loop

- AI systems now generate hypotheses, candidates, and rankings.
- But many pipelines are still stitched together with scripts and notebooks.
- That breaks reproducibility, governance, and learning speed.

Building models is no longer enough; operating them is the real bottleneck.

What "Ad Hoc" Looks Like

```
model_v1 → notebook_eval → spreadsheet_metrics → manual notes → model_v2
   |              |                |                   |
 unclear      non-repeatable   no lineage          no audit trail
```

- Which exact run produced this claim?
- Why did performance shift this week?
- Is this gain real, or split leakage / randomness?

Systematic Improvement Loop

The point is not one good run; the point is a reliable learning system.

2

W&B Tools and How They Help

Models + Weave + Evaluations as the operating layer

W&B Models: Versioning and Lineage

- Register base models, tuned checkpoints, and aliases.
- Track datasets, configs, and artifacts per run.
- Promote with explicit criteria (not intuition).

```
model: bioreason-pro
alias: pilot-disease-go-v3
parent: bioreason-pro-base
training_data: disease-human-go-v2
eval_gate: temporal_holdout_go_v1 ≥ target
```

Weave: Traces for Agents and Model Calls

- Capture each call, intermediate step, latency, and output.
- Debug failed predictions with full context.
- Compare behavior across versions and prompt templates.

```
@weave.op
def go_function_agent(sample):
    # trace input → retrieval → reasoning → output
    return run_agent(sample)
```

Evaluation: Measuring Real Improvement

```
same split + same metrics + same protocol → fair comparison
```

- Evaluate by biologically meaningful metrics and slices.
- Track confidence intervals and temporal holdout behavior.
- Decide promotions with explicit eval gates.

3

What We’re Doing: Pilot Plan

BioReason-Pro pilot on disease-related human proteins for GO function prediction

- Objective: improve GO prediction quality.
- Constraint: **no architecture changes**.
- Focus: data strategy + tuning strategy + evaluation rigor.

Ablation Grid (Planned)

```
Axes:
1) Data scope
   - generic proteins
   - + disease-related human proteins
   - + curated hard cases

2) Training strategy
   - baseline (no tune)
   - SFT
   - RL (GRPO-style variants)

3) Prompt / context strategy
   - base prompt
   - ontology-aware prompt
   - retrieval-augmented context
```

Run each cell with the same eval suite for apples-to-apples comparisons.

Tuning Strategies We Are Testing

Baseline

Current BioReason-Pro, no tuning.

SFT

Supervised fine-tuning on curated disease-protein examples.

RL Variant A

Reward prioritizes GO correctness + calibration.

RL Variant B

Reward adds robustness penalties for known failure modes.

Execution Plan

```
Week 1: freeze baselines + finalize eval split
Week 2: run SFT ablations
Week 3: run RL variants + reward sensitivity
Week 4: compare, select, and package candidate model
```

- Every run logged in W&B with linked artifacts.
- Promotion candidate requires passing predefined eval gates.
- Keep all decisions traceable for technical and compliance review.

4

Results + Live Demo

Slides for summary, W&B UI for evidence

Results Snapshot (Replace with latest)

TBD	TBD	TBD	0
delta Fmax	delta AUPRC	best strategy	arch changes

- Show confidence intervals on final version of this slide.
- Keep metric definitions visible in backup slides.

Live W&B Demo: What I Will Show

1. **Run comparison dashboard** (baseline vs SFT vs RL variants)
2. **Trace deep dive** for one success and one failure case
3. **Model registry lineage** from base model to candidate
4. **Eval report + promotion gate** decision

Goal: show that every claim in slides is backed by inspectable evidence.

Closing Message

- Drug discovery AI needs model innovation **and** operating discipline.
- A systematic loop turns one-off experiments into cumulative progress.
- W&B tooling helps teams trace, evaluate, tune, and govern at production pace.
- Our pilot asks a focused question and tests it with reproducible evidence.

If it cannot be reproduced and explained, it cannot be trusted.

Questions

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